Infrastructure Governance & Compliance Manual

This technical manual documents the deterministic thresholds integrated into the ShineGuard Hulo platform. These boundaries are derived from international electrotechnical standards (IEC), road lighting codes (EN), and cybersecurity best practices (OWASP/NIST).

1. Electrical & Telemetry Thresholds

Metric	Nominal Range	Warning Level	Critical Trip	Compliance Standard
Voltage	210V - 240V	< 200V / > 250V	< 180V	***IEC 60038***
Temperature	20°C - 55°C	> 65°C	> 85°C	***Industrial Grade***
Current (A)	0.01A - 2.0A	> 2.5A	> 5.0A	***IEC 60598-1***
Ambient Light	10 - 1000 lux	< 10 lux	0 lux	***EN 13201***
Humidity	10% - 60%	> 80%	> 95%	***IP65/67 Standard***

• **Voltage Rationale:** Following IEC 60038, we allow a ±10% tolerance, but trigger a critical trip at 180V to prevent Smitched-Mode Power Supply (SMPS) current spikes that lead to thermal runaway in LED drivers.

• **Thermal Management:** The 85°C limit is the standard case temperature redline for industrial semiconductors. Operating above this point causes "Thermal Aging," significantly reducing the hardware lifespan.

• **Electrical Safety (Current):** A 5.0A trip point is implemented for short-circuit protection. Since a 50W LED draws ?0.2A, this buffer handles initial "Inrush Current" while protecting against fire hazards.

2. Cybersecurity & Governance

Rule Name	Operational Limit	Standard Alignment	Security Justification
API Velocity	60 req / min	OWASP API7:2023	DDoS Mitigation
Auth Expiry	15 mins Idle	NIST SP 800-63B	Hijacking Defense
Bulk Lockout	5 Min Window	High-Assurance IAM	Multi-factor Re-Auth
Log Chain	HMAC-SHA256	FIPS 180-4	Non-Repudiation

• **Velocity Control (OWASP):** Prevents automated "Scraper" bots and Brute-Force attempts by limiting requests to 1 per second, ensuring system resources remain available for telemetry.

• **Cryptographic Integrity:** By chaining logs using HMAC-SHA256, we provide a mathematical proof of "Non-Repudiation," ensuring that no activity log can be deleted or modified without detection.

3. Asset Lifecycle Thresholds

Lifecycle Event	Reference Value	Maintenance Category
Service Interval	12 Months / 4k Hrs	Preventative (EN 13306)
Health Threshold	> 5 Critical Errors	Condition-Based Maintenance
Resolution Target	< 24 Hours	SLA Compliance (ISO 9001)
Retirement Age	5 Years / 25k Hrs	Total Cost of Ownership (TCO)

• **Preventative Longevity (EN 13306):** Routine servicing at the 4,000-hour mark prevents hardware "Fatigue" and reduces overall emergency repair costs by approximately 40%.

- **Workforce Accountability (SLA):** The 24-hour MTTR (Mean Time To Repair) target is the primary Service Level Agreement (SLA) metric used to audit city maintenance responsiveness.

-- END OF OFFICIAL MANUAL --